IGCSE 0607 Practice Questions — E4.2 — Combined Questions & Markscheme

Gradient of linear graphs

Total Questions: 9 — **Total Marks:** 50 Marks

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  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

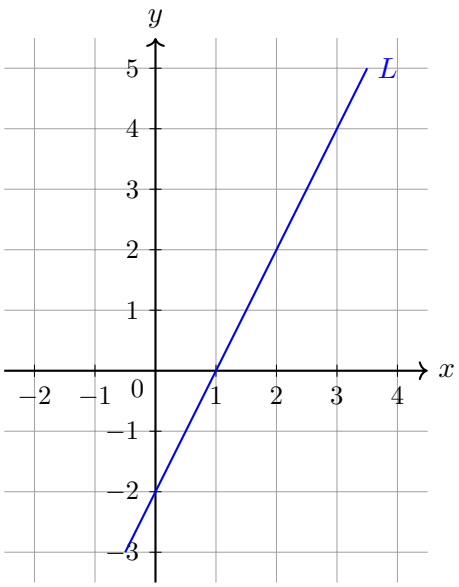
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1 Points A and B have coordinates $(2, -3)$ and $(6, 5)$ respectively.

Work out the gradient of the line AB.

..... **[2]**

2 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the gradient of line L.

..... **[2]**

(b) Write down the coordinates of the y-intercept of line L.

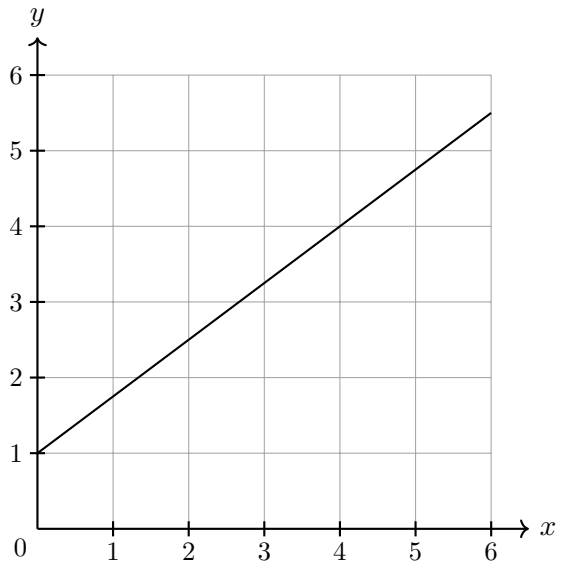
(..... , ...) **[1]**

3 The points A and B have coordinates $(-2, 8)$ and $(4, -4)$ respectively.

(a) Work out the gradient of the line AB.

..... **[2]**

(b) A straight line L is drawn on the grid.
Find the gradient of line L.



..... **[2]**

4 **(a)** Point P has coordinates $(2a, 3a)$ and point Q has coordinates $(5a, a + 4)$.

Find an expression for the gradient of the line PQ in terms of a.
Give your answer in its simplest form.

..... **[2]**

(b) The line PQ is perpendicular to a line with gradient 3.
Work out the value of a.

$a =$ ..... **[3]**

(c) The points $A(-1, 2)$, $B(3, 8)$ and $C(c, 23)$ lie on a straight line.
Work out the value of c.

$c =$ ..... **[3]**

5 The line L_1 passes through the points $P(-2, 3a)$ and $Q(4, a+6)$.

(a) Find an expression for the gradient of L_1 in terms of a.
Give your answer in its simplest form.

..... **[3]**

(b) The line L_2 has equation $y = 3x - 5$.
Given that L_1 is parallel to L_2, find the value of a.

$a =$ ..... **[2]**

(c) Using your value of a, find the coordinates of the midpoint of PQ.

(..... , ...) **[3]**

6 The points A, B and C have coordinates $(1, 3)$, $(4, 9)$ and $(8, y)$ respectively.

(a) The points A, B and C lie on the same straight line.
Work out the value of y.

$y =$ ..... **[2]**

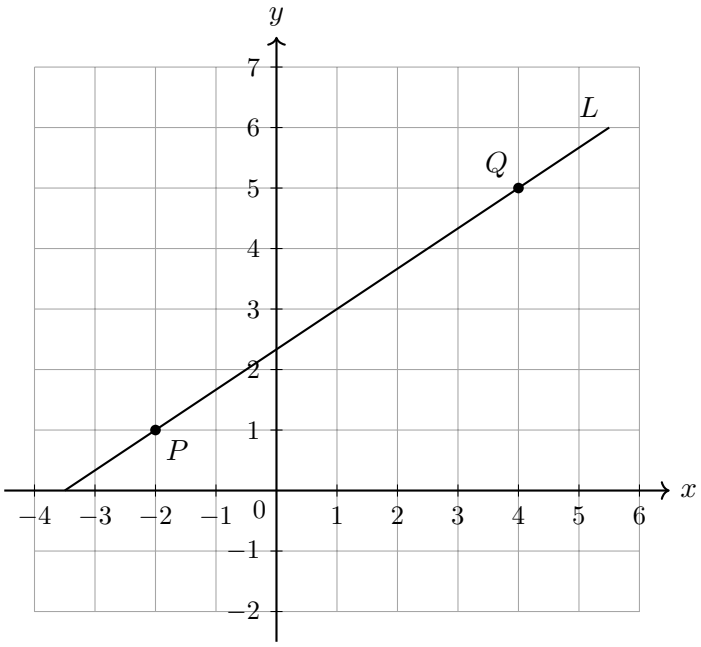
(b) A different point D has coordinates $(p, p^2 - p - 3)$.
The gradient of the line AD is 2.
Work out the possible values of p.

$p =$ ..... , $p =$ ..... **[4]**

(c) A line L has gradient m.
The line L passes through the points $(3, m)$ and $(m + 1, 12)$.
Find the two possible values of m.

$m =$ ..... , $m =$ ..... **[4]**

7 **(a)** The diagram shows a straight line L passing through the points $P(-2, 1)$ and $Q(4, 5)$.
Find the gradient of line L.



..... **[2]**

(b) A different line passes through the points $(3, k)$ and $(8, -2)$.
The gradient of this line is -1.4.
Calculate the value of k.

$k =$ ..... **[3]**

8 Point A has coordinates $(-3, 5)$ and point B has coordinates $(4, -9)$.

(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Point C has coordinates $(k, 1)$.
The gradient of the line AC is equal to the gradient of the line AB.
Find the value of k.

$k =$ ..... **[3]**

9 Point P has coordinates $(k-3, 2k)$ and point Q has coordinates $(2, 5)$.

(a) The gradient of the line PQ is -3.
Work out the value of k.

$k =$ ..... **[3]**

(b) Using your value of k from part (a), find the coordinates of the midpoint of PQ.

(..... , ...) **[2]**

1 $\frac{5-(-3)}{6-2}$ or equivalent **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 **(a)** Evidence of using $\frac{\text{change in } y}{\text{change in } x}$ with two correct points from the graph, e.g., $\frac{4-(-2)}{3-0}$ **[M1]**

2 **[A1]**

(b) $(0, -2)$ **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** Uses gradient formula: $\frac{-4-8}{4-(-2)}$ or $\frac{8-(-4)}{-2-4}$ **[M1]**

-2 **[A1]**

(b) Identifies coordinates of two clear points on the line, e.g., $(0, 1)$ and $(4, 4)$ **[M1]**

$\frac{3}{4}$ or 0.75 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** $\frac{(a+4)-3a}{5a-2a}$ or equivalent **[M1]**

$\frac{4-2a}{3a}$ **[A1]**

(b) States or implies gradient of PQ is $-\frac{1}{3}$ **[B1]**

$\frac{4-2a}{3a} = -\frac{1}{3}$ or $3(4-2a) = -3a$ **[M1]**

4 **[A1]**

(c) Gradient $AB = \frac{8-2}{3-(-1)}$ or $\frac{6}{4}$ or $\frac{3}{2}$ **[M1]**

$\frac{23-8}{c-3} = \frac{3}{2}$ or $\frac{23-2}{c-(-1)} = \frac{3}{2}$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 8]*

5 **(a)** $\frac{(a+6)-3a}{4-(-2)}$ **[M1]**

$\frac{6-2a}{6}$ **[M1]**

$\frac{3-a}{3}$ or $1-\frac{a}{3}$ **[A1]**

(b) $\frac{3-a}{3} = 3$ oe **[M1]**

$a = -6$ **[A1]**

(c) $P = (-2, -18)$ and $Q = (4, 0)$ **[M1]**

$\left(\frac{-2+4}{2}, \frac{-18+0}{2}\right)$ **[M1]**

$(1, -9)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

6 (a) Calculates the gradient of AB as $\frac{9-3}{4-1} = 2$ and equates to gradient of BC or AC, e.g., $\frac{y-9}{8-4} = 2$ **[M1]**

$y = 17$ **[A1]**

(b) Sets up the gradient equation: $\frac{(p^2-p-3)-3}{p-1} = 2$ **[M1]**

Simplifies to form a 3-term quadratic equation: $p^2 - p - 6 = 2(p-1)$

$\mathrm{p}^2 - 3p - 4 = 0$ [M1]

Factorises $(p-4)(p+1) = 0$ or uses the quadratic formula correctly **[M1]**

$p = 4$ and $p = -1$ **[A1]**

(c) Sets up the gradient equation: $\frac{12-m}{(m+1)-3} = m$ **[M1]**

Multiplies by the denominator: $12 - m = m(m-2)$ **[M1]**

Rearranges into a 3-term quadratic: $m^2 - m - 12 = 0$ **[M1]**

$m = 4$ and $m = -3$ **[A1]**

[Back to Question] *[Total Marks: 10]*

7 (a) $\frac{5-1}{4-(-2)}$ or equivalent **[M1]**

$\frac{2}{3}$ **[A1]**

(b) $\frac{-2-k}{8-3} = -1.4$ or equivalent **[M1]**

$-2 - k = -7$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 5]*

8 (a) $\frac{-9-5}{4-(-3)}$ or $\frac{5-(-9)}{-3-4}$ **[M1]**

-2 **[A1]**

(b) $\dfrac{1-5}{k-(-3)} = -2$ oe **[M1]**

$-4 = -2(k+3)$ or $-2k-6 = -4$ **[M1]**

$k = -1$ **[A1]**

[Back to Question] *[Total Marks: 5]*

9 (a) Forms a correct equation for the gradient: $\dfrac{5-2k}{2-(k-3)} = -3$ OR $\dfrac{2k-5}{k-3-2} = -3$ **[M1]**

Correctly simplifies the denominator and equates: $5-2k = -3(5-k)$ **[M1]**

$k = 4$ **[A1]**

(b) Evaluates coordinates of P as $(1, 8)$ and substitutes into midpoint formula: $\left(\dfrac{1+2}{2}, \dfrac{8+5}{2}\right)$ **[M1]**

$(1.5, 6.5)$ **[A1]**

[Back to Question] *[Total Marks: 5]*